

# MATB24H3

## Exercise Questions

Student Name: Your Name

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2.1.1. Consider the set  $X = \mathbb{Q} \setminus \{0\}$  and the map  $\star$  is depicted by  $a \star b = \frac{a}{b}$ . Is  $\star$  a binary operation?

2.1.2. Is  $(\mathcal{P}(\mathbb{N}), \cap)$  a monoid? what is the identity element? How about  $(\mathcal{P}(\mathbb{N}), \cup)$  for any arbitrary set  $X$ ?

2.2.1. Suppose  $d$  is square-free, so there is no integer  $m$  such that  $m^2 = d$ . Show that  $(\mathbb{Q}(\sqrt{d}), +, \cdot)$  forms a field. What happens if  $d = m^2$  for some  $m$ ?

2.2.2. Now, can we extend a finite field in the same way we extended an infinite field? If so, give an example.

2.2.3. Prove that  $0_{\mathbb{F}}$ ,  $1_{\mathbb{F}}$  and the additive inverse element  $-a$  of any element  $a \in F$  are unique.

2.2.4. Prove  $(-a)(-b) = ab$  for all  $a, b \in \mathbb{F}$ .

2.2.5. Prove that the triple  $(\mathbb{Z}_p, + \bmod p, \cdot \bmod p)$  is a field  $\iff p$  is a prime. Use the fact that if two integers  $m, n$  are relatively prime, then there exist integers  $r$  and  $s$  such that  $mr + ns = 1$ .

2.2.6. Prove that the 2-tuple  $(S, \circ)$ , where  $S$  is the set of symmetries of a square and  $\circ$  is the composition operation, does indeed form a group. Moreover, find the symmetry group of a rectangle.

2.2.7. Characterise all idempotent elements of  $(\mathcal{M}_{2 \times 2}(\mathbb{R}), +, \cdot)$ .

2.2.8. Prove that  $\mathcal{C}(\mathcal{R})$  is a subring of any ring  $(\mathcal{R}, +, \cdot)$ .

2.2.9. Let  $(\mathcal{R}, +, \cdot)$  be a ring and suppose that  $\alpha \in \mathcal{R}$  has a multiplicative inverse. Moreover, suppose that  $\alpha \in C(\mathcal{R})$ . Is  $\alpha^{-1} \in C(\mathcal{R})$ ?

2.2.10. Does the set of functions with infinitely many roots form a subring of  $\mathcal{F}(\mathbb{R})$ ? Justify your answer.

2.2.11. Prove the following statements:

- (a) There exist rings  $(\mathcal{R}, +, \cdot)$  with an identity element  $1_{\mathcal{R}}$  such that  $1_{\mathcal{R}} \notin S$  for some subring  $S$ .
- (b) [TERM TEST QUESTION] There may exist a ring  $\mathcal{R}$  with an identity element  $1_{\mathcal{R}}$  and a subring  $S \subseteq \mathcal{R}$  such that  $S$  has its own identity element  $1_S$ , where  $1_S \neq 1_{\mathcal{R}}$ .

2.2.12 Determine the zero and the identity elements of the polynomial ring  $(\mathcal{R}[x], +, \cdot)$  where  $\mathcal{R} = \mathcal{C}(\mathcal{M}_{2 \times 2}(\mathbb{R}))$ .

2.2.13 True/False. Justify:

1. Let  $\mathbb{F}$  be any field. Suppose that  $\alpha$  and  $\beta$  are two elements of an extension field  $\mathbb{K}$  of  $\mathbb{F}$ . Then  $\mathbb{F}(\alpha)$  and  $\mathbb{F}(\beta)$  are always distinct extension fields of  $\mathbb{F}$ .
2. Given any set  $X$ , there is no operation  $\cdot$  that can be defined on  $[X]^{\leq \|X\|}$  that makes it a group.
3. Given a monoid  $(\mathcal{S}, *)$  there exists a binary operation  $\cdot$  on  $\mathcal{S}$  such that  $(\mathcal{S}, \cdot)$  no longer admits an identity element. If so, what assumption did you need to make?
4.  $(\mathbb{R}, \cdot)$  forms a group.
5.  $(\mathbb{R}, +)$  forms a group with the identity element being 1.
6. Suppose  $(\mathbb{F}, +, \cdot)$  is the field. Then  $(\mathbb{F}, +)$  and  $(\mathbb{F}, \cdot)$  both form a group.